

# Animal Science and Industry

# Animal health

# 12

Animal health refers to the physiological well-being of an animal. It is a broader topic than the presence or absence of an infectious disease. Animal health is certainly affected by a variety of infectious diseases, but it is also affected by noninfectious diseases, animal density, stress, boredom, nutrition, availability of water, temperature, cleanliness, attention and care, shelter, and other factors.

The normal, healthy animal is content, alert, and productive. A laying hen will have bright eyes, an attentive and alert manner, the head held high, and absence of skin, head, or leg blemishes. Healthy cattle on pasture will graze contentedly and intermittently. They will have a smooth hair coat, walk with ease, and generally will be with or close to others of the herd. A healthy young lamb will nurse aggressively, will play, run, and jump with other lambs, and alternately stretch out in the sun to rest. A riding horse in good health will hold the head high, be inquisitive, carry a smooth hair coat, seek affection, be lean and trim, and walk or run with a free and clean stride.

The observing manager will note these traits and others, and quickly note deviations in appearance or behavior – listlessness, rough hair coat, drooping head, dull eyes, arched back, slow movement, separation from the herd or flock – that suggest health is impaired.

Subclinical illnesses or low levels of parasitism that cause decreased feed intake and gain may not manifest themselves in an obvious manner, however, unless feed consumption and weight gains are recorded.

Upon completion of this chapter and related sections in other chapters, the reader should be able to (1) list a series of factors that influence animal health; (2) differentiate among them in regard to the intensity of effect that negative factors may have on animal comfort and productivity; (3) define the word "disease" and differentiate between infectious and noninfectious diseases and provide examples of each; (4) describe the immunization process and explain why vaccination is practiced; (5) describe conditions in livestock and poultry that disclose to a herd or flock manager that health is impaired; and (6) explain the role of the veterinarian.

## 12.1 ECONOMIC LOSSES

It is estimated that losses caused by animal diseases in the United States annually amount to as much as 15 percent of the potential gross income from animals and animal products. Ten to 20 percent of all calves, lambs, and pigs die between birth and weaning; 2 to 4 percent of all chicken broilers and turkeys die before they reach market weights, and laying hens typically have a mortality rate of about 1 percent per month.

Mortality, however, is only a part of the monetary cost. Other costs are vaccines and chemicals used to prevent or control diseases, veterinary expense, reduced productivity of diseased animals, cost of federal and state disease-eradication programs, and value of product condemned at processing plants. In 1980, it was estimated the disease mastitis (inflammation of the mammary gland) cost dairy producers an average of $170 per cow in costs of treatment and lost milk, yet the disease does not cause mortality. The brucellosis control program in the United States cost over $123 million in 1979. Routine vaccination is deemed necessary to prevent diseases such as Newcastle disease and infectious bronchitis in chickens, erysipelas in swine and turkeys, rabies in dogs, shipping fever in cattle, and enterotoxemia in lambs, at considerable cost to producers. Table 12-1 lists types of losses attributable to a single disease, and benefits derived by disease prevention measures.

Antibiotics and other chemicals are fed to most young monogastric animals to prevent certain bacterial diseases. Anthelmintics are used to eliminate or control intestinal worms, and coccidiostats are fed to young chickens to prevent the protozoan disease, coccidiosis. The necessity of using these products increases production cost.

Some diseases are given special consideration because they affect both domestic animals and humans. These diseases, called "zoonoses," include bovine tuberculosis, brucellosis (called undulant fever in humans), psittacosis (parrot fever), rabies, trichinosis, Q fever, salmonellosis, and others. Certain of these have been brought under control through federal testing programs. See Section 12-11. Others are controlled by sanitation, preventive medication, and thorough cooking of food products. Revenue lost due to consumers' fear (usually unwarranted) of contracting disease through animal products is not measurable.

Table 12-1

BENEFITS DERIVED FROM CONTROL OF MAREK'S DISEASE THROUGH VACCINATION

| *Source of benefits* | *Annual amount ($ millions)* |
|---|---|
| Reduced leukosis broiler condemnations | 27.3 |
| Reduced other broiler condemnations | 6.3 |
| Reduced broiler mortality | 5.6 |
| Reduced broiler breeder mortality | 5.0 |
| Reduced egg-type chicken mortality | 15.4 |
| Reduced condemnations of egg-type chickens | 0.1 |
| Improved feed utilization | 3.2 |
| Increased egg production | 105.5 |
| Total | 168.4 |

Source: H. G. Purchase and E. F. Schultz, Jr., "The Economics of Marek's Disease Control in the United States," *World's Poultry Sci. J.* 34 (1978): 198-204.

## 12.2 DISEASE—DEFINED AND CLASSIFIED

The term "disease" may be defined in livestock and poultry production as an impairment to normal body health, an illness or a malady.

Some diseases are caused by specific, living agents, such as parasites, bacteria, protozoa, or viruses, and these are called infectious diseases. Some infectious diseases are contagious — they can be transmitted to other animals directly or indirectly. Examples include coccidiosis, anthrax, and leptospirosis.

Noninfectious diseases may include certain genetic abnormalities, nutritional deficiencies (anemia), metabolic diseases (pregnancy toxemia), poisoning (lead), and other maladies.

The term "disease" is not defined in the same way by all. The above definition may be considered broad. Some restrict the term to maladies that are infectious or to those that are contagious, and would list the other type of impairments mentioned in this chapter simply as parasite infestations, genetic anomalies, nutritional deficiencies, stress syndrome, and so forth.

The purpose of this chapter and related sections of other chapters, however, is to discuss the total topic of animal health and the groups of factors in livestock and poultry production that may impair health and, therefore, animal comfort and productivity.

## 12.3 PREDATORS AND INJURIES

Confinement of animals to feedlots, swine production units, drylot dairies, and broiler and layer houses minimizes predator risk. Wildlife predators

remain a problem, however, to the sheep rancher and to a lesser extent, to the cattle rancher. In fact, predators may be a more serious problem to ranchers in the 1980s than before imposed constraints on predator control.

The coyote is the most significant predator in ranching areas,* but the bobcat, bear, and cougar have also been known to cause problems. Bands of domestic dogs may also be predators, both in ranching areas and where ewe flocks or lambs are maintained near residential areas. Losses in sheep flocks are significant. Total loss to predators in range states in 1974 ranged from 7.5 to 16 percent of the stock sheep (mature animals) and from 3.8 to 21.7 percent of the range lambs.

Wire cuts in horses, especially among those kept in small pens, and teat injuries in the cow – from frostbite, from being cut by coarse grass or a weed, or from being stepped on where cows are bedded in open sheds – are among the more common injuries.

Cannibalism in poultry and tail and ear biting in swine are discussed in Section 13.13.

## 12.4 NUTRITIONAL DEFICIENCIES

In earlier years, when many swine were routinely produced on pasture, when layers ranged over a farmstead and could consume grass and insects, and when poultry for meat were raised on grass or ranged over the farmstead, nutritional deficiencies were less common or, because of less advanced knowledge at that time, less observed.

As confinement of nonruminant animals became more common, nutrients were available only through provided rations, and as genetic selection and other techniques were used to increase productivity per animal unit, more nutritional deficiencies were observed and identified.

Deficiencies may result from several causes: (1) a low level of the nutrient such as an essential mineral, in the soil, resulting in forage or grain being especially low in that nutrient; (2) a prepared ration that is too low in a key nutrient; (3) impaired absorption of a nutrient; (4) incomplete or blocked metabolism of a nutrient in the body cells; (5) abnormal requirement for a nutrient, due perhaps to incomplete metabolism, but possibly due to some environmental factor; or (6) failure of the animal to synthesize a nutrient that is normally synthesized in the digestive tract or body cells (and thus considered nonessential in the diet).

Some of the more common nutritional deficiencies observed in livestock and poultry are listed in Table 12-2.

Nutritional deficiencies are generally avoided by provision of nutritionally complete rations, formulated according to available research information. Where soils may be deficient in iodine, iodized salt is made available or mixed with concentrate. Where locally grown and lower-cost feeds are the

*Maurice Shelton and Dale Wade, *Animal Industry Today* 2, No. 1 (1979): 4.

**Table 12-2**

**COMMON DEFICIENCY DISEASES IN DOMESTIC ANIMALS**

| *Element or nutrient* | *Species affected* | *Disease condition* |
|---|---|---|
| Vitamin A | Primarily cattle and sheep | Weak calves at birth |
| | All species to some degree | Night blindness |
| | | Keratitis |
| | | Reproduction problems |
| | | Congenital defects |
| Vitamin D | All species | Rickets |
| | | Joint disease |
| | | Osteomalacia |
| Vitamin K | All species | Hemorrhagic condition |
| Calcium[a] | All species | "Milk fever" (cows) |
| | | Osteoporosis |
| | | Rickets |
| Phosphorus | All species | Pica |
| | | Poor growth |
| | | Infertility |
| Selenium and vitamin E | All species | Muscular pathology |
| | | Retained placenta |
| | | Encephalomalasia in poultry |
| Magnesium | Primarily cattle | Grass tetany |
| | | Transportation tetany |
| Zinc | Primarily swine | Parakeratosis |
| Manganese | Primarily cattle | Infertility |
| | | Skeletal deformities |
| | Poultry | Skeletal deformities |
| Cobalt | Primarily cattle and sheep | Digestive problems |
| | | Loss of body condition |
| Iron | All species | Anemia |
| Copper | Sheep | Enzootic ataxia |
| | Swine | Anemia |
| Iodine | All species | Goiter |
| | | Partial or complete alopecia |

[a]Calcium deficiency is closely associated with calcium/phosphorus ratios and availability of vitamin D.

major ration ingredients but may be deficient in a nutrient, the ration is designed to include a rich source of that nutrient, even though that ingredient may be more costly or needs to be imported. An example is corn-soybean meal rations for swine and poultry in the Midwest. Such a ration is usually low in the amino acids lysine and methionine, so blood meal, fishmeal, or some other rich source of lysine and methionine usually is included in the ration.

Manufactured or mined nutrients – usually vitamins and minerals – are often added.

It is important to recognize there may be two *levels* of deficiency. An acute or severe deficiency would likely cause the visible symptoms described

in Table 12-2. In the early 1980s, however, such deficiencies were rare in well-managed herds and flocks. Deficiencies that are borderline and that are difficult to perceive or identify may simply prevent health and production from being at the *optimum* level.

Considerable research is directed toward the latter, and producers in many cases will provide a "safety factor," an extra amount of an apparently critical nutrient in the ration. This is especially common where knowledge appears incomplete, where the animals may be under considerable stress, or where the nutrient content of feed ingredients may vary. At the same time, excessive levels should not be added. Extremely high levels of vitamin A will cause epithelial problems, for example, and excess levels of certain minerals will interfere with absorption or utilization of other nutrients. In the case of calcium and phosphorus, the ratio of the two is considered as important as the actual levels.

## 12.5 METABOLIC DISORDERS

Metabolic disorders are often difficult to separate from nutritional deficiencies. The blockage of a metabolic pathway may cause a deficiency at the body cell, even though the ration is entirely adequate. The fact that increasing the nutrient content of the diet may eliminate the symptom (by offsetting the metabolic blockage) often adds to the belief a dietary deficiency may exist.

Common metabolic disorders of domestic animals are summarized in Table 12-3.

## 12.6 TOXINS AND POISONS

For this discussion, *toxins* are defined as those harmful chemicals of animal cell or bacterial origin or production. *Poisons* are defined as those chemicals harmful to animals where the source is a growing or harvested plant, or the material has been chemically or biologically manufactured. In practice the terms are used interchangeably, so it is well to discuss them as a group.

Illnesses initiated by toxins are called toxemias. Toxins can be generally divided into two categories: (1) antigenic toxins, produced by bacteria or other microbiological life, and (2) metabolic toxins, produced by animal body cells.

Antigenic toxins are usually proteins and trigger an antibody reaction in the animal. Body temperature goes up and clear symptoms of illness result. Enterotoxemia, caused by *Clostridium perfringens* type D in feedlot lambs, is an illness caused by an antigenic toxin.

Ketosis in the cow is caused by metabolic toxins. Improper carbohydrate metabolism, or lack of sufficient carbohydrate, results in massive catabolism

Table 12-3

COMMON METABOLIC DISEASES OF DOMESTIC ANIMALS

| *Species* | *Disease* | *Causative factor* |
|---|---|---|
| Horse | Lactation tetany of mares | Calcium deficiency (early lactation) |
| | Azoturia | Sudden increase in muscular work or high carbohydrate diet |
| Cattle | Ketosis | Hypoglycemia |
| | "Fat cow syndrome" | Excessive conditioning especially during dry period |
| | "Milk fever" | Hypocalcemia |
| | "Transport tetany" | Calcium-magnesium deficiency |
| | "Grass tetany" | Magnesium deficiency |
| | "High mountain" or "brisket disease" | Chronic hypoxia related to high elevations |
| | Postparturient hemoglobinuria | Phosphorus deficiency? |
| | "Whole-milk tetany" in calves | Magnesium deficiency |
| | Photosensitization | Liver pathology – may be related to plant toxins |
| | Porphyria | Congenital |
| Swine | Hypoglycemia (piglets) | Postpartum deprivation of milk |
| | "PSS" (porcine stress syndrome) | Heritable defect |
| Sheep | Pregnancy toxemia | Impaired metabolism of carbohydrates or advanced pregnancy stress |

of stored fats to supply energy. End products of this catabolism – ketones – accumulate and are toxic to the animal.

High levels of nitrites in rapidly growing plants, plants whose growth may be interrupted by an early frost, or plant materials harvested during a rapid growth stage may cause poisoning. Nitrite is an intermediate step in protein manufacture and large quantities would be present at a given time in a rapidly growing plant. Nitrite is toxic; protein is not. Frost or harvesting prevents the plant from converting the nitrite to protein.

A number of inorganic and organic compounds have been incriminated in livestock toxicity or poisonings, including:

| *Inorganic* | | *Organic* |
|---|---|---|
| Lead | Fluorine | Hydrocyanic acid (prussic) |
| Mercury | Sodium chloride | Nitrate or nitrite |
| Arsenic | Selenium | Oxalate |
| Copper | Zinc | |
| Molybdenum | Sulphur | |

Especially high levels of selenium are contained in the soils of parts of south central South Dakota, and in the grasses and grains that grow there, resulting in malformed hooves and horns on cattle and reduced prolificness. Cases of lead poisoning in animals that lick lead-based paints are common.

Fluorine has been a serious problem where it has settled out on grass leeward from aluminum manufacturing plants or plants where rock phosphate is converted to the feed ingredient, dicalcium phosphate.

Anthelmentics (worm-controlling agents) such as carbon tetrachloride, hexachlorethane, phenothiazine, nicotine, and cadmium, are safe when administered under optimum dosage and conditions but may be poisonous at high doses or when administered to debilitated animals.

Animal or plant insecticides, such as chlorinated hydrocarbons, organophosphatic compounds, and rotenone, as well as herbicides, rodenticides, and wood preservatives, have been incriminated as animal poisons.

## 12.7 PARASITES AND PROTOZOA

Livestock and poultry are subject to infestation by both internal and external parasites.

Common internal parasites include, by host species:

*Cattle*

Stomach worms
Intestinal round worms
Intestinal tapeworm
Lungworms
Liver fluke
Grubs

*Sheep & Goats*

Stomach worms
Intestinal round worms
Intestinal tapeworm
Lungworms
Nose bots

*Horse*

Stomach worms
Intestinal worms
Tapeworms

*Swine*

Stomach worms
Intestinal worms
Lungworms
Swine kidney worm
Trichina

*Poultry*

Coccidia
Tapeworms
Intestinal round worms
Capillary worms
Gapeworms

The principal intestinal parasites affecting domestic animals are roundworms and tapeworms. Adult roundworms produce eggs which pass out with feces. If temperature and moisture are adequate the eggs develop to larvae. After ingestion by a suitable host, the larvae (immature worms) burrow into the intestinal mucosa, causing irritation and damaging the absorptive surfaces of the intestine. Eventually mature roundworms emerge into the intestine lumen and produce eggs, completing the life cycle. In addition to damage caused by the immature worm, adult worms may become so numerous that they cause nutritional deficiencies in the host or even cause blockage of the intestines.

Tapeworms are flat, segmented worms with indirect life cycles – part of the life cycle occurs within one or more specific *secondary* hosts (insects, arthropods, slugs, etc.).

Within the digestive tract of the animal, the head and neck sections of the immature worm are embedded permanently in the intestinal lining. Segments grow out of the neck section in a chainlike fashion and the worm may reach several feet in length. Each segment has both a male and female gonad and, as the segments mature, thousands of eggs are produced. As the egg-filled segments break off, they pass out of the animal's body with the feces. Some are ingested by a secondary host and the eggs undergo changes that make them infectious for another secondary host or the primary host animal.

The fact that the head section of the tapeworm remains permanently embedded in the intestinal lining, means that eliminating tapeworms is usually more difficult than eliminating roundworms. It also suggests that the appearance of tapeworm segments in the feces of treated animals does not assure that the head and neck sections have been removed.

The most common external parasites in all species of livestock and poultry are lice, mites, and flies.

Lice may damage their host by biting or blood sucking, depending upon species. Some mites, including most poultry mites, live on the skin surface, feeding at night and leaving the host during daylight. However, the tiny Northern Fowl Mite, the most common external parasite in chickens, spends its entire life on the bird's body.

Other mites burrow into hair follicles, skin glands, or skin tissue per se. Mange of dogs, swine, and other mammals is caused by the burrowing type of mites.

Heel flies cause much economic loss to cattle producers. Although the adult flies neither bite nor sting, they greatly annoy cattle. These large flies attach eggs to hairs on the heels or legs of cattle. Developing larvae penetrate the animal's skin, migrate through the animal's body, and after five to six months appear under the skin of the back. Each larvae, called a grub, makes a breathing hole through the skin. Eventually, the grub emerges, drops to the ground, and pupates to form young adult flies. Hides of infected animals have reduced value; annual losses caused by this parasite are estimated at well over $100 million.

Horn flies and stable flies are blood-sucking and annoying insects that cause reduced growth rate in beef cattle and reduced milk production by dairy cattle. Other ectoparasites of importance include fleas, ticks, horseflies, and mosquitoes.

Blood-sucking external parasites may also be vectors of disease (mosquitoes, for example, transmit equine encephalitis virus). Others may neither bite nor suck blood but transport microorganisms from one host to another (face flies carry bacteria that cause pinkeye in cattle).

Microscopic examination of skin scrapings and fecal material is often necessary to achieve a complete and accurate diagnosis of internal or external parasite or protozoan infestations.

Though some chemicals are used as preventives, animal management systems that include periodic and rigid cleaning of facilities, good nutrition and care, rapid treatment of injuries, and rotation of animals among different pastures or lots are generally effective in reducing parasite burdens. Each of these contributes to breaking the life cycle of the parasite or increasing the resistance of the animals to infestation.

In the case of some parasites, prevention systems are more specific. Horn flies and face flies are credited with the spread of eye infections in cattle, especially the bacterial disease, pinkeye. While cattle are on pasture, subjected to eye stresses such as wind, sunlight, pollen, weeds, and tall grass, pinkeye infections are common. In Indiana and Kansas beef herds, insecticide-impregnated plastic tags attached to the ears of cattle have effectively reduced the presence around the head of horn flies and face flies. The incidence of pinkeye was also sharply lower in the Kansas herds, and the weaning weight of calves was 15 to 20 pounds greater.*

Treatments are generally internally or externally administered chemicals that serve as poisons to the parasite or protozoa. Obviously such chemicals should be used according to established recommendations, and the manager should be especially alert to state or federal restrictions on use, in the case of laying hens or dairy cows, or where affected animals may soon be slaughtered for meat.

## 12.8 TYPES OF INFECTIOUS AGENTS

The terms "infectious" and "contagious" are often used loosely and interchangeably. For this discussion, an infectious agent is defined as one that is capable of causing an infection in the body. A contagious agent is an infectious agent that may be readily transmitted from one body to another. The term "communicable" is considered a synonym to the term "contagious." Localized abscesses of animals are often infections caused by the bacterium *Corynebacterium pyogenes*. However, that bacterium does not readily spread

*Eugene N. Francis, Kansas Cooperative Extension Service, private correspondence, November 9, 1981; and R. E. Williams et al., *J. Animal Sci.* 53 (1981): 1159.

from the affected animal to another animal. By contrast, anthrax, caused by the bacterium, *Bacillus anthracis*, is both infectious and highly contagious; it is transmitted readily from one animal to another.

Disease-producing agents are sometimes discussed as *primary* versus *secondary* agents. For example, "lumpy jaw" in cattle may be initiated by a plant awn breaking the mucosa within the mouth and then the bacterium *Actinomyces bovis*, a common inhabitant of the bovine mouth, establishes itself and causes the infection termed "lumpy jaw." The plant awn would be the primary agent, *Actinomyces bovis* the secondary agent.

Infectious agents range in size from multicelled fungi or large single-celled protozoa to viruses so small that they can be observed only with an electron microscope.

*Viruses* reproduce inside living cells of the host. They lack cytoplasm and utilize the metabolic system of the host cell to replicate. The host cell loses its ability to divide, eventually degenerates, and the virus particles which are then released invade other cells.

Many viruses are host-specific and may be tissue-specific. For example, hog cholera, formerly one of the most serious swine diseases, occurs naturally only in swine. Eastern equine encephalitis virus, pathogenic to both humans and horses and capable of infecting birds, primarily affects the host's central nervous system tissue. Many respiratory diseases such as equine rhinopneumonitis, canine distemper, virus pneumonia of pigs, and infectious bronchitis in chickens are caused by viruses.

Some virus diseases are transmitted from host to host only by specific vectors (e.g., equine encephalitis by mosquitos) while others such as hog cholera may be spread by direct contact between animals or by indirect contact (feed, water, boots, vehicles). Control measures include isolation, strict sanitation, elimination of vectors, exclusion of visitors from production units, and vaccination programs.

In the case of widely disseminated virus diseases vaccination of susceptible animals is routine. See Table 12-4 for a sample vaccination program for replacement pullets.

Bacteria are microscopic, single-celled organisms which reproduce by binary fission and which vary greatly in size, shape, pathogenicity, and host specificity. The zoonotic diseases tuberculosis, brucellosis, and salmonellosis are caused by bacteria belonging to the genera *Mycobacterium*, *Brucella*, and *Salmonella*, respectively.

Bacteria damage their hosts' tissues primarily by the production of toxins. Some bacteria (e.g., staphylococci and streptococci) produce and release toxic substances as by products of their metabolism. These toxins are referred to as exotoxins and symptoms appear in the animal soon after infection occurs. Other bacteria, such as salmonellae, produce endotoxins - cellular components that are released to cause damage to the host only when the bacterial cells lyse or disintegrate. Symptoms, therefore, appear in the animal later than do symptoms caused by exotoxins.

Bacterial infections are prevented by use of antibiotics or other chemo-

Table 12-4

SAMPLE VACCINATION SCHEDULE FOR REPLACEMENT PULLETS

| *Age of pullets* | *Vaccine given* |
|---|---|
| 1 day | Marek's disease |
| | Newcastle disease |
| | Infectious bronchitis |
| 10 days | Newcastle disease |
| 6 weeks | Newcastle disease |
| | Infectious bronchitis |
| 8 weeks | Laryngotracheitis |
| 10 weeks | Fowl pox |
| 12 weeks | Laryngotracheitis |
| 14 weeks | Newcastle disease |
| | Infectious bronchitis |

cals, vaccination, and testing programs that provide for elimination of infected individuals.

Protozoa are single-celled animals, microscopic in size, but larger than bacteria. The most common protozoan diseases of domestic animals are in a group referred to as "coccidiosis" and caused by protozoan parasites called coccidia. Typically coccidia invade the epithelial cells of the digestive tract and exist as intracellular parasites throughout most of their life cycle. The coccidia often replicate asexually within the cells of the host's intestinal tract, cause rupture of the host cells, and invade other cells for continued replication. After a number of asexual generations, male and female gametes are formed and united within the host's cells, forming oocysts. The oocysts are eliminated with feces and, given adequate conditions of temperature and moisture, sporulate (divide) outside the animal's body. The sporulated oocysts can remain viable and infectious for several days or weeks. If oocysts are ingested by a new host during that time, the disease may spread.

Coccidia are very host-specific. Thirteen species of coccidia affect cattle and 11 affect swine, but none of the cattle species are infectious for swine; chickens have 9 species of coccidia belonging to a single genus (*Eimeria*), none of which affect turkeys or other animals.

Coccidiosis causes extensive damage to the lining of the digestive tract. The disease is prevented by including drugs called coccidiostats in the ration. When outbreaks of the disease occur, anticoccidial drugs are usually administered in the drinking water.

*Fungi* are single-celled or multicellular plants of a low order of development.

Best known among filamentous fungi are species of *Tricophyton* which cause "ringworm" in pets as well as farm animals. Sometimes seen in humans, it can be acquired by handling infected pets.

Aspergillosis in birds is caused by a filamentous fungus. This disease

results from inhalation of the fungal spores and their germination and growth in the respiratory system. The disease in young birds is often called "brooder pneumonia" because of the pneumonia-like symptoms and its occurrence during the early life of the bird.

Histoplasmosis, a systemic fungal disease of canines and humans, is an infectious, noncontagious disease. The organism is usually inhaled with dust particles and often causes infection of the lungs.

Fungal diseases are difficult to treat and are best controlled by preventing exposure of susceptible animals. The feeding of moldy feedstuffs and use of moldy litter or bedding should be avoided.

## 12.9 IMMUNITY AND VACCINATION

Immunity is a state of resistance in the animal to a disease-producing agent. It may be inherited or acquired.

Though it has been observed for decades that some animal species are not bothered by certain infectious diseases, the scientific study of inherited resistance to livestock and poultry diseases has been intensely pursued only in recent decades. There is much evidence that tendency toward or resistance to mastitis and cancer eye in cattle are inherited. (See the heritability estimates in Table 18-1.) Commercial poultry breeding programs now include, in many cases, the object of achieving some level of genetic immunity to certain diseases, such as Marek's disease. The opportunities in other species may be high, but because of the lower prolificness and longer generation time in those species, less has been learned and what may be learned will have less rapid implementation.

In practice, most resistance is acquired, either by exposure of the young animal to the existing diseases or by some form of artificial exposure – vaccination. A newly born or hatched animal is immediately exposed, in the natural environment, to a multitude of microorganisms and viruses – in the bedding or litter, in the soil, on the teat or tongue of the dam, in the water, and so forth. Many of these are capable of causing infection, especially should the young animal be susceptible – in a weak condition or in a stressful situation.

But most newly born or hatched animals have naturally provided disease-fighting mechanisms. All have in the blood a relatively high circulation level of immunoglobulins, nonspecific disease-fighting proteins.

Preformed antibodies are passed across the placenta from the dam to the fetus in humans, dogs, cats, and guinea pigs, but not in pigs, cattle, sheep, and horses. In the latter mammals, the milk produced the first few days after birth of the offspring, colostrum, is especially high in immunoglobulins, other protein (used in production of other specific antibodies), and fat. (Composition of colostrum was shown in Table 6.5.) Thus, the young animal will likely acquire passive immunity to specific agents. It is very important that calves nurse their dams within a few hours after birth, since the ability

of the intestine to absorb the large immunoglobulins conferring passive immunity rapidly diminishes.

The newly hatched chick or poult is nourished by the egg yolk, which is also high in immunoglobulins.

Development of immunity by *exposure* of the young animal to an existing disease explains several phenomena in livestock and poultry production. Animals born or hatched and kept during their early weeks in especially hygienic conditions or in totally new facilities may become ill when moved to more normally managed, previously used facilities where there is a normal population of potential disease-producing agents. The same often occurs when stock is moved from one country to another, or one latitude to another, and a markedly new and different population of organisms is encountered.

In the case of certain diseases, the causative organism or virus is rare enough that the young animal doesn't normally encounter it (and therefore doesn't develop a natural immunity) and/or so virulent that when encountered it often causes serious illness and death. For such cases, scientists have developed systems to create immunity in the animals that may later be exposed. The following paragraphs describe the basis for these systems, begun about a century ago.

The French chemist and biologist, Louis Pasteur, developed in the 1870s and 1880s a serum that would provide resistance to anthrax in an animal that had received the serum. The serum had been separated from the blood of an animal that had been infected with the disease, had developed an antibody resistance to it, and had recovered. Test sheep were first injected with serum, then were injected with an infectious culture of the anthrax organisms. Those sheep that had first received the serum remained well. A second group, which had not received the serum, became ill and died when injected with the culture of anthrax organisms.

Anthrax is a virulent (quick-acting and strong) disease. Apparently the serum provided temporary and "passive" immunity until the infected animals could develop their own antibodies (active immunity) to an extent sufficient to prevent the disease.

In the ensuing century, a series of discoveries and developments have resulted in a variety of vaccination techniques for all species and for a variety of infectious diseases. Most have been directed toward triggering the production of immunoglobulins or other disease-fighting agents in the susceptible animal. During the earlier years, the technique that Pasteur used – (1) inject serum from a recovered animal, (2) inject a culture of the disease-causing organism, and (3) depend on the animal to achieve permanent resistance – was followed in most cases. Species that were not as susceptible to the disease and/or that would yield large volumes of serum were kept and used in serum manufacturing plants. Horses and swine were especially useful. Cultures of the live agent – usually called "bacterin" or "virus" were prepared in manufacturing laboratories.

In some instances, the disease was so virulent that there was much risk in using a live culture of disease-causing organisms or viruses on farms and

ranches. A broken bottle or an accidental thrust of the syringe plunger could contaminate the soil or facilities for decades, or directly infect animals that had not received serum. Eventually, forms of attenuated (reduced virulence) or dead cultures that would trigger the antibody reaction but not cause severe illness were developed. This and other developments resulted, in some cases, in the injection of serum being unnecessary.

After this refinement was achieved, it took many years, of course, before production units or areas where live bacteria or live virus had been used could be rid of all apparent live material. In the case of hog cholera, special educational programs urged use of attenuated or dead virus. As a state or area approached a low incidence of the disease, laws forbidding the use of a live virus were passed in order to eventually achieve a cholera-free designation. This designation enhanced the market for breeding stock and feeder pigs, and suggested a safer environment for swine production.

Especially for poultry, where individual injection would be laborious and costly, vaccines were developed that could be administered via the water or as an aerosol.

Similar developments with other diseases and other species could be cited.

## 12.10 REPRODUCTIVE AND RESPIRATORY PROBLEMS

Reproductive and respiratory problems are major concerns in many livestock and poultry production units in which a variety of infectious agents, nutritional deficiencies, and other factors may be involved. They illustrate well the present complexity of animal health problems when animals are under considerable production stress, animal density may be high, and movement of people, animals, and feed is wide-ranging and rapid.

Infectious reproductive diseases among bovine include brucellosis, vibriosis and leptospirosis, all of which can be carried and transmitted by sires. These diseases can be effectively prevented by vaccines. But vitamin A or phosphorus deficiencies, low energy or protein levels, high environmental temperatures, and other factors can also diminish prolificness in either the dam or sire. The number of possible interactions is high, and the difficulty of sorting out likely causes of reproductive problems is apparent.

Respiratory problems in all species are complex. Though specific infectious agents are often identified in nasal cultures, a mixture of infectious agents often exists. Research has demonstrated that apparent virulence of an agent may be influenced by presence or absence of others, by dust, by the ration, and by stress. In fact, even though there are specific respiratory diseases, such as infectious bovine rhinotracheitis (IBR), many respiratory infections are often referred to simply as a "respiratory disease complex."

Horses may encounter equine viral rhinopneumonitis (EVR), swine often have influenza, and airsacculitis is common in poultry. It is emphasized that stress on the individual animal is usually a major contributing factor to establishment or the consequences of these diseases.

## 12.11 ERADICATION AND REPOPULATION PROGRAMS

Gradual elimination of hog cholera by elimination of the use of live virus continued vaccination with modified vaccines, and eventual absence of cases was described in Section 12.9. Other diseases have been eradicated or reduced in incidence to the point that they are no longer of major economic importance in the United States.

Foot and mouth disease and contagious pleuropneumonia are examples. The incidence of tuberculosis and brucellosis have been significantly reduced. In the case of tuberculosis, the major technique used was routine testing at state or federal expense of all breeding cattle, and slaughter of all infected animals. Though some indemnity was paid the owner, there was often severe economic loss. In a similar brucellosis program, USDA indemnity payments are supplemented in most states by state or county indemnity payments.

In general, northern states have been more aggressive in developing and carrying out brucellosis control and eradication programs. Incidence was sharply reduced during the 1960s and early 1970s, especially in the northern states. At the lower incidence level most of the cases detected are in mature cows at slaughter plants. These cows are tagged, the owner identified, and the total herd from which they came is tested.

In the late 1970s severe drought in some of the northern states forced sharp reductions in cow herd size. Normal rainfall in later years resulted in many cows and heifers being brought in from southern states where brucellosis is more common. In one northern state the number of known brucellosis-infected cow herds increased from 4 to 40 in one year. State inspection and surveillance programs were increased. All imported heifers and cows were inspected, held in quarantine for 30 days, and inspected again. Carriers of the disease went directly to slaughter.

Eradication programs require complete cooperation by producers and others, both within and between states, plus inspection and quarantine provisions at national boundaries.

Where outbreaks of specific and especially virulent or economically significant diseases such as foot and mouth disease occur, elimination and proper disposal (by burning, burial, and/or cooking to produce tankage) of all susceptible animals within the area is essential. Following cleanup and disinfection, "sentry" animals are usually placed in the area. If, after a period of time and thorough examination, no evidence of the disease exists in the sentries, repopulation proceeds.

## 12.12 TYPICAL HERD-HEALTH PROGRAMS

Most herd or flock managers follow a specific and planned health plan, based on the life cycle of the animals or based on the calendar. An example, a herd health plan for a swine breeding herd is shown in Fig. 12-1.

All health programs depend heavily on complete records on individuals or on groups handled together.

Scheduled visitation and consultation by a veterinarian, perhaps under a contract that calls for a payment per animal or per hour, is often an integral part of a herd health plan. A program for a dairy client in the Midwest may call for monthly dairy farm visits and include pregnancy testing, checking for mastitis or parasites, providing routine vaccinations, dry cow mastitis treatment, and foot trimming. A typical dairy herd health program is shown in Fig. 12-2.

The objective of all herd health programs is prevention of disease and maintenance of normal health and high production in the herd or flock.

**Fig. 12-2.** A dairy herd health program adapted from one used at Ohio State University. Note biweekly herd examinations by a veterinarian. (Ohio State University)

DAIRY HERD HEALTH PROGRAM

*Newborn Calves*

Hand feed first milking of colostrum.
Treat navel with iodine disinfectant.
Remove extra teats and dehorn at 1 to 2 months of age.

*Vaccination*

Leptospirosis – twice yearly for all cattle over 3 months.
Infectious Bovine Rhinotracheitis, Bovine Viral Diarrhea, and Parainfluenza – 3 vaccines at 8 to 12 months of age.

*Mastitis Control*

Dip teats in disinfectant post milking.
Test all cows monthly.

*Reproduction*

Herd examination by veterinarian every 2 weeks to include
Pregnancy exam of cows 30 to 45 days postbreeding.
Examination of cows 0 to 15 days after calving and at about 30 days post calving.
Examination of cows not bred after 3 or 4 services.
Examination of cows not seen in heat (anestrus) by 60 days postpartum.
Removal of exposed portions of retained placenta from postpartum cows on first regularly scheduled examination, or earlier if cow has fever or is off feed.

*Parasite Control*

Worm heifers during first grazing season—3 and 6 weeks after going on pasture.
Use approved insecticide bags on all stock continuously for control of lice.

*Emergencies*

Dystocias (difficult birth)—call veterinarian or experienced herdsman.
Aborted fetuses and dead animals—arrange for necropsy by veterinarian as soon as possible.
Sick or lame animals—call veterinarian immediately.

## 12.13 THE ROLE OF THE VETERINARIAN

The veterinarian functions in livestock and poultry production systems as a member of a flock- or herd-health team. That team includes the flock or herd manager, all workers, nutritionists, and perhaps other specialists. The veterinarian has concentrated training in animal diseases, medicine, surgery, and diagnosis (Figs. 12–3, 12–4).





**Fig. 12–3.** A modern veterinary field hospital in the Kansas Flint Hills, top. Note cattle-working and sorting pens. Right photo shows cattle-restraining chute inside hospital. Bottom photo is of a mobile unit for handling cases on farms and ranches. (Kansas State University)



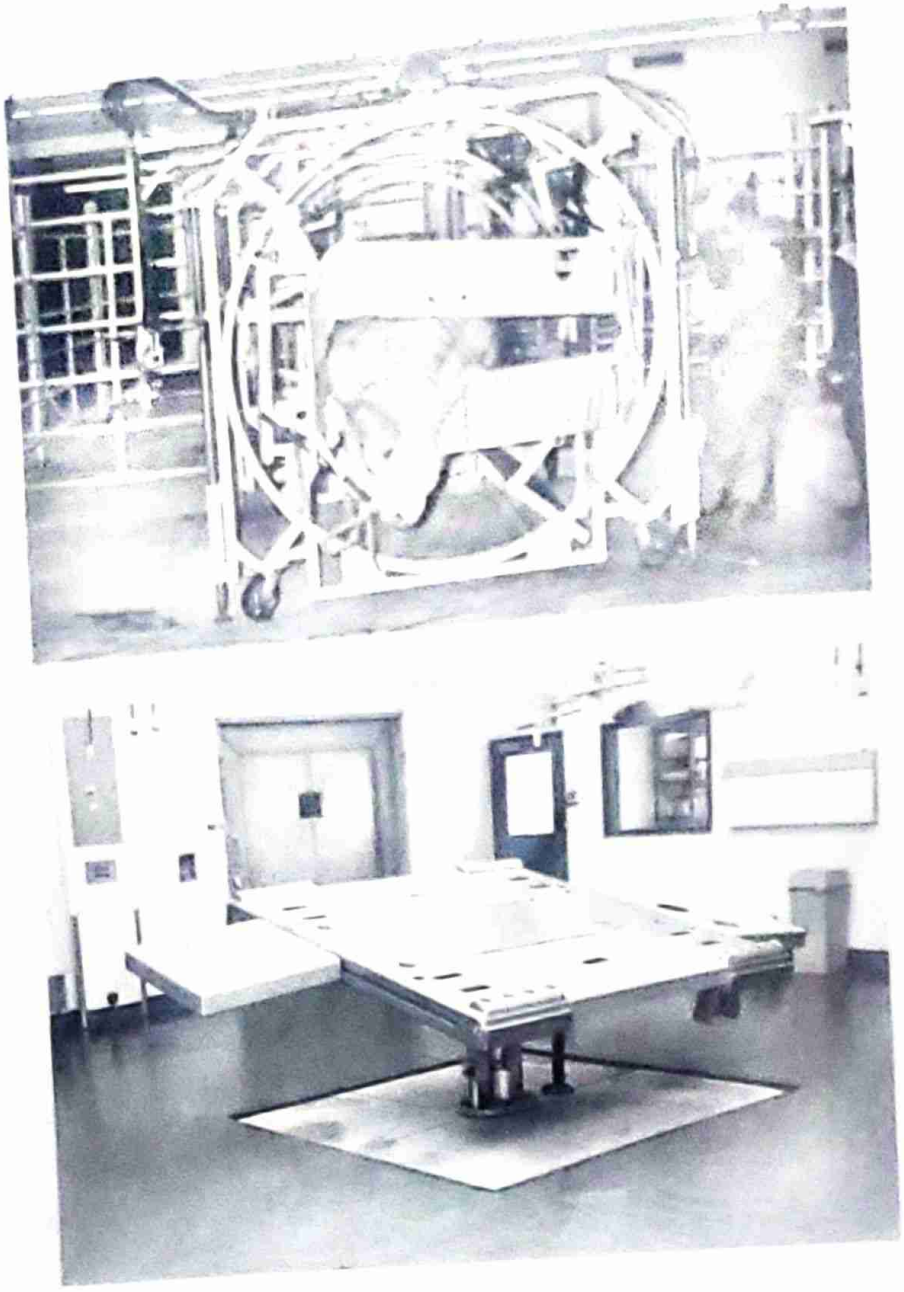


**Fig. 12–4.** Large animal-restraining table with 360° rotation potential, top, and tilting surgical table, bottom, in a College of Veterinary Medicine. Clinical cases referred by practicing veterinarians provide training opportunities for veterinary students. (Kansas State University)

In pursuit of specific diagnosis, a veterinarian may depend on microbiologists, parasitologists, metabolic or reproduction physiologists, pathologists, biochemists, and scientists in other disciplines. All who work with animals should be familiar with the characteristics of the normal animal, including body temperature, pulse, and respirations. These are listed in Table 12-5. Normal temperatures will vary 0.5 to 1.0° C in 24 hours, with the minimum in the morning and the maximum in the afternoon.

High environmental temperatures, will often cause elevated body temperature; so will exercise. Infections commonly cause a fever (elevated temperature). Toxemias or shock will often cause a body temperature drop.

Pulse and respiration rates generally increase as the body temperature increases or as a result of excitement or exercise. The pulse rate may be determined by palpating the ventral side of the tail or facial arteries of the cow, the facial arteries of the horse, and the inner thigh of sheep and goats. It is difficult to obtain an accurate pulse in the pig by palpation.

Although temperature and pulse and respiration rates are important indicators of health, deviations from the normal behavior, such as listlessness,

Table 12-5

AVERAGE TEMPERATURE AND PULSE AND RESPIRATION RATES OF COMMON DOMESTIC ANIMALS

| *Species* | *Temperature (rectal)* | *Pulse rate (per min)* | *Respirations (per min)* |
|---|---|---|---|
| Cow | 101.5°F (38.6°C) | 60 | 20 |
| Horse | 100.0°F (37.8°C) | 35 | 12 |
| Pig | 102.5°F (39.2°C) | 60–100 | 8–18 |
| Sheep | 102.5°F (39.2°C) | 60–90 | 12–20 |
| Goat | 102.5°F (39.2°C) | 60–90 | 12–20 |
| Poultry | 106.0°F (41.1°C) | 250–350 | 17–27 |

decreased feed intake, slow movement, or unusual posture are clues to impaired health. Physical traits can best be observed from a reasonable distance and in the animal's natural environment. Following are examples of clues that may help a veterinarian and the flock or herd manager diagnose an illness, the essential step before treatment can be initiated.

Examination of individual animals, to determine the nature of the illness and possible causes, usually begins by examination of the eyes, nostrils, and ears. Excessive discharges from the eyes or nostrils suggest respiratory problems. Excess salivation may suggest immobility of the tongue, central nervous disturbances, dental problems, or blockage of the esophagus. The attitude (alignment) of the head and neck is often altered with difficult breathing; an animal with pneumonia will often extend the head and neck.

Alterations from the normal respiration rate, rhythm, and depth suggest a respiratory disease such as pneumonia. Thorax and brisket edema (swelling with fluid) is commonly associated with a circulatory disturbance and respiratory distress.

An arched back is commonly associated with pain in the ventral thorax such as that caused by traumatic reticuloperitonitis (hardware disease) where a foreign body (nail, wire, etc.) has penetrated the reticulum.

The abdomen and its symmetry and fullness often provides clues. Bloat will initiate marked swelling of the dorsal, left abdomen; gastrointestinal blockage, on the other hand, will usually cause a general abdominal distention. Gauntness indicates the cow likely has not eaten for at least several days. A combination of gauntness and tucking up of the abdomen might indicate abdominal soreness.

Infections of the vulva, penis, and scrotum may cause swelling of varying degrees. Blockage of the urinary tract in the male, such as with urinary

calculi, may be detected by checking the penis for the presence of calculi or stones. Vaginal discharges, other than those associated with the estrous cycle, may indicate infection of the genital tract.

Swelling of the mammary gland often indicates mastitis, while a flaccid udder in a producing cow suggests a systemic infection or metabolic disturbance.

The feces should be observed closely. The lack of fecal passage or constipation may indicate a digestive disturbance or may be the result of a systemic disease accompanied by high fever. Diarrhea may suggest inflammation of the intestine or stomach. Bright red blood in the feces suggests coccidiosis or rectal injury. Blood discharged from gastric ulcers and/or the anterior of the intestine would cause very dark feces. Excess mucus may result from intestinal inflammation and glistens in the feces.

Examination of the limbs for abnormalities is important, especially in the horse. Watching a horse walk or trot will disclose impairments. Founder will slow and change the rhythm of the gait. So will fractures and dislocations of bones, joint strains, and abscesses on the feet. Soreness of the thorax (chest cavity) causes a changed gait and abnormal positioning of the elbows when at rest.

The veterinarian who services livestock and poultry producers and others concerned with animal health are supported in their work by one or more diagnostic laboratories administered by the state university college of veterinary medicine or department of veterinary science in most states, or in some by the executive branch of the state government. The laboratory subjects tissues, live animals, or dead animals to routine testing or to detailed diagnostic scrutiny to ascertain or confirm causes of illness or death, and issues timely reports by phone or otherwise so that treatment may begin or the management regime may be adjusted.

Colleges of veterinary medicine also operate a clinic or hospital, which provides learning opportunities for veterinary students and handles routine and emergency cases in the geographic area of the college and special or unusual cases from throughout its own and neighboring states. College, diagnostic laboratory, clinic, and extension veterinary faculty present conferences and short courses for veterinarians and producers, and publish bulletins and other materials on animal health.

# The egg as a product

# 34

This chapter is devoted to the egg as a food item. The egg as an instrument of reproduction is discussed in Chapter 14.

The egg has quality attributes – shell texture and shape, shell and yolk colors, and flavor – that are established by genetic and environmental factors. Some of these attributes are subject to change after the egg is laid and subjected to handling, storage, and processing. Sections that follow discuss the role each of these factors plays in egg quality.

The egg is a source of valuable nutrients; it can be a part of the main course, or used in salads, desserts, or snacks. The egg is subject to classification and grading by the USDA or private companies. It is an instrument of commerce, to be bought, sold, and promoted. It is a source of financial profit. This chapter includes sections devoted to these topics, and upon completion the reader should be able to (1) sketch an egg and identify its major parts; (2) contrast the physical and nutritional characteristics of the egg albumen and yolk; (3) list the USDA egg grades and the major factors that determine the grade of an egg; (4) list the major factors that influence egg quality; (5) describe how eggs are normally handled, after they leave the hen, to ensure maintenance of quality; (6) list a series of products that contain processed eggs; and (7) describe the trend in U.S. per capita egg consumption.

## 34.1 CHARACTERISTICS AND ATTRIBUTES

Figure 34-1 shows the parts of an egg and also a cross section of the shell. The yolk develops, while attached to the hen's ovary, as a potential source of nutrients for a developing embryo, were the egg to become fertilized. The yellow color is due to fat-soluble pigments, mostly xanthophylls transferred from the digestive tract via the blood stream and usually supplied by such feeds as alfalfa meal or yellow corn.

The albumen (white) includes all material between the vitelline yolk membrane and the inner shell membrane, and is secreted to surround the yolk in successive sections of the hen's oviduct. Composition of the yolk and white are shown in Table 34-1. The shell membranes are lacelike protein materials.

The inner layer of the shell, formed first, is apparently two or three layers of tiny round calcified pebbles (mammilla) deposited on the outer "shell" membrane. These are then covered and cemented together, except for certain pores (Fig. 34-2), by a second layer, a calcareous spongy material. The third, outer, shell layer is a thin protein material, which tends to seal the shell. This outer layer is called the cuticle.

Shell shape and strength are important. An egg that is too long or has too large a circumference may be crushed, or squeezed and cracked, while being packed or moved in containers. One that is especially short and round tends to roll faster, and can become cracked or can crack others on gathering belts or in graders. Uniformity of shape and texture help appearance.

Shell texture and thickness are major quality attributes. The pores which admit oxygen for embryonic development in a fertilized egg are a detriment to a table egg; the simultaneous loss of carbon dioxide permits oxidation and quality deterioration. Some have observed that brown shells, on the average, are stronger than white shells; if so, this may be related to the number of

**Fig. 34-1.** The parts of an egg. (USDA)

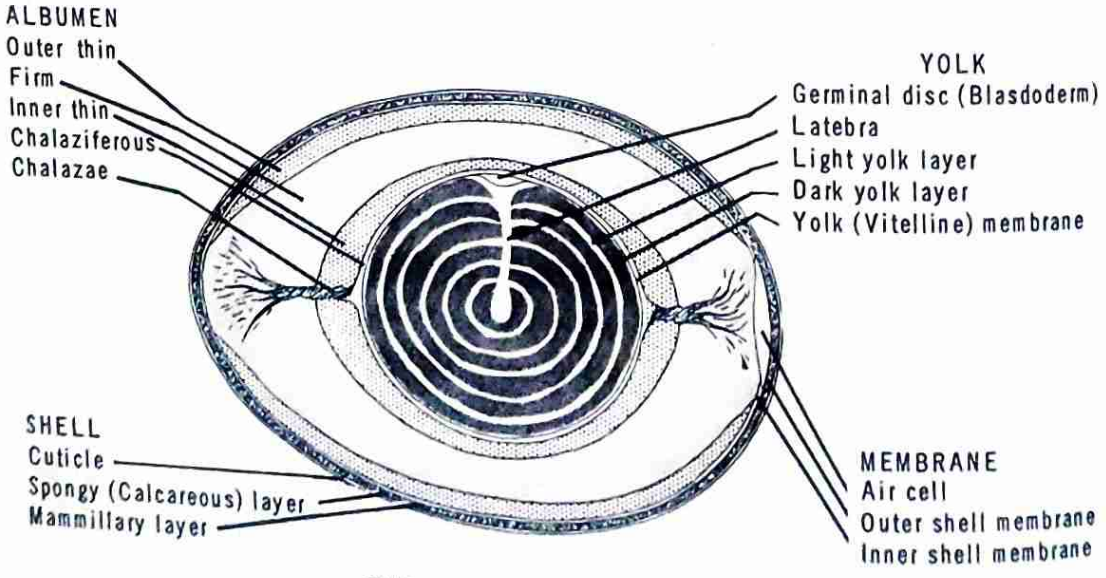


Table 34-1

AVERAGE COMPOSITION OF WHOLE EGG AND COMPONENTS AS PERCENT OF TOTAL

| | *Percent by weight* | *Percent* | | | |
|---|---|---|---|---|---|
| | | *Dry matter* | *Protein* | *Fat* | *Ash* |
| Whole egg | 100 | 34.5 | 11.8 | 11.0 | 11.7 |
| (2.61)[a] | | 90 | 20.8 | 28.7 | 30.5 |
| White | 58 | 12 | 11.0 | 0.2 | 0.8 |
| (7.50) | | 90 | 82.5 | 1.5 | 6.0 |
| Yolk | 31 | 52 | 17.5 | 32.5 | 2.0 |
| (1.87) | | 90 | 30.3 | 56.2 | 3.5 |
| | | *Calcium carbonate* | *Magnesium carbonate* | *Calcium phosphate* | *Organic matter* |
| Shell | 11 | 94.0 | 1.0 | 1.0 | 4.0 |

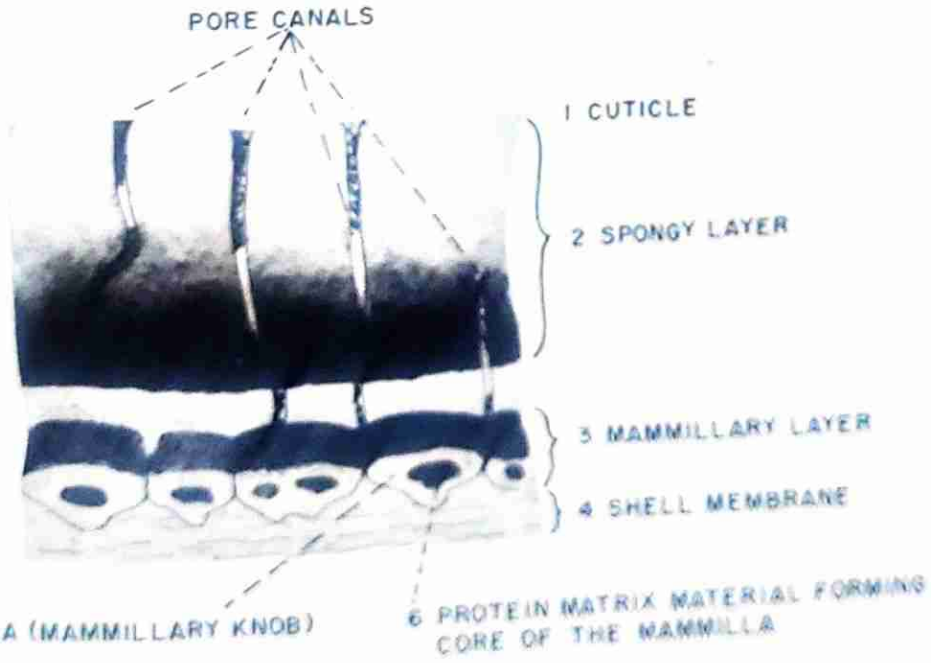
Source: Data adapted from FSQS, USDA, *Agricultural Handbook No. 75*, 1972.

[a]Figures in parentheses are the pounds in natural state equivalent to one pound of air-dry material (90 percent dry matter). This allows comparison with other foods such as those in Table 26-5. In that table composition data for "eggs" excludes the shell.

eggs laid – the fewer eggs laid, the more calcium per shell. Layers of brown shelled eggs usually lay fewer eggs.

Breeders of egg-type chicken strains measure shell strength by (1) micrometer measurement of shell cross section, (2) force required to break the shell, and (3) specific gravity. A weak or porous shell will admit more air, causing the egg to float in water, so the specific gravity is lower.

Fig. 34-2. A magnified cross section of the egg shell. (USDA)



The proportion of thick albumen, and the degree to which the albumen remains thick until the egg is used, is a major quality trait. Consumers prefer thick albumen; when an egg is broken for frying or poaching, an egg with a thick albumen retains a small circumference and stands up high. This is measured by breaking out an egg onto a flat surface and measuring the height of the albumen. The height is expressed *in relation* to egg weight as "Haugh units." A thin albumen denotes lack of freshness, though albumen quality may also be influenced by other factors. Eggs laid late in the hen's production cycle usually have lower Haugh unit scores. Some pullets lay eggs with relatively thin albumen, especially when environmental temperature is high (see Section 34.4).

Consumers like a clear albumen, a high yolk, definite and uniform yolk color (there is variation among consumers in depth of yolk color preferred), and chalaza which are not prominent. Chalaza are spiral bands of thicker albumen, contained within the albumen or white, and which help to maintain the yolk in its position. Freedom from "blood spots" (traces of blood resulting from capillary rupture at the time of ovulation) or "meat spots" (these are usually traces of tissue from the hen's ovary or oviduct) is also an important trait.

Eggs are especially nutritious. The albumen is very high in protein and the protein is comprised of a desirable balance of amino acids. The yolk contains fat and is also high in protein; fat-soluble vitamins A, D, E, and K; and the elements iron, phosphorus, sulfur, copper, potassium, sodium, magnesium, calcium, chlorine, and manganese.

## 34.2 CLASSES AND GRADES

Eggs are classed according to size (ounces of weight per dozen) and cleanliness, and graded according to standards of quality by USDA graders or by handlers, perhaps using USDA criteria as guides.* They are usually also sorted by color. Grades and weight classes are provided for consumer, wholesale, and procurement purposes. Only consumer classes and grades are described here. The wholesale classifications are essentially the same, except that percentage tolerances are provided because large lots are handled. Table 34-2 summarizes weight classes.

Eggs are subjectively graded by mass scanning, usually called candling. They are passed over a lighted background so that shell soundness and certain interior quality features can be checked. Individual eggs may be "hand candled" by holding the large end up to a box opening, behind which is a bright light.

*Material in this section describes grades and classes effective October 1981. It was adapted from *Agricultural Handbook No. 75*, published by FSQS, USDA, as modified and reported in the *Federal Register*, August 4, 1981.

Table 34-2

U.S. WEIGHT CLASSES
FOR CONSUMER GRADES OF FRESH EGGS

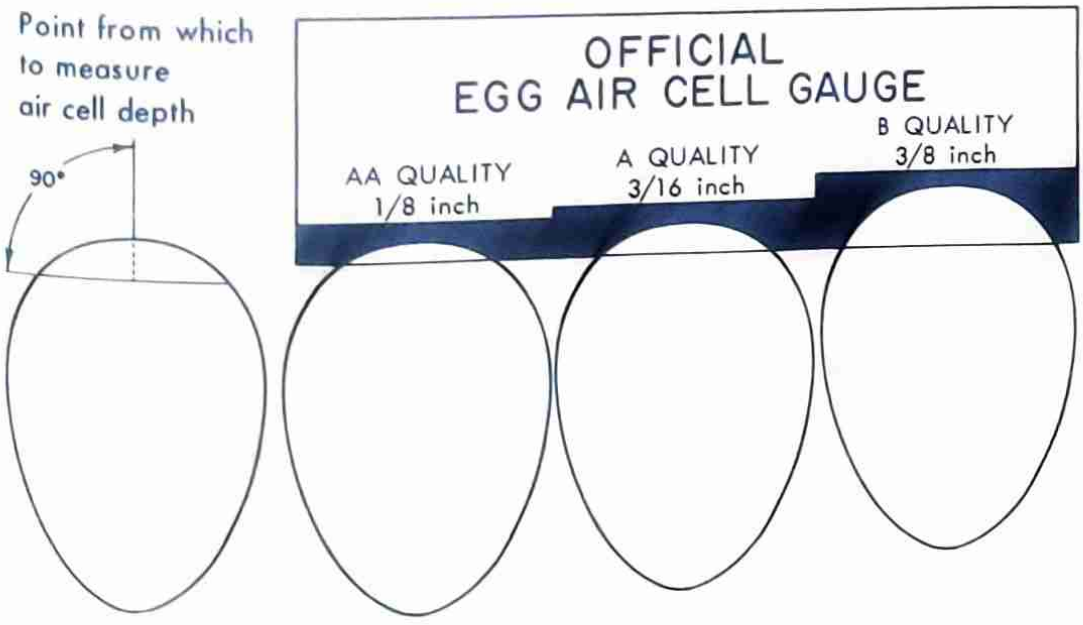
| *Size or weight class* | *Minimum net weight per dozen (oz)* | *Minimum net weight per 30 dozen (lb)* | *Minimum weight for individual eggs at rate per dozen (oz)* |
|---|---|---|---|
| Jumbo | 30 | 56 | 29 |
| Extra large | 27 | 50 $\frac{1}{2}$ | 26 |
| Large | 24 | 45 | 23 |
| Medium | 21 | 39 $\frac{1}{2}$ | 20 |
| Small | 18 | 34 | 17 |
| Peewee | 15 | 28 | — |

Source: FSQS, USDA, *Agricultural Handbook No. 75*, 1972.

One can assess the size of air cell, distinctness of yolk shadow outline and shape of yolk, and can detect yolk or albumen defects or germ development by rotating the egg before the candling light.

The air cell forms after the egg is laid, because the white and yolk cool and shrink, and it may continue to increase in size during storage due to evaporation of moisture from the liquid contents (Fig. 34-3). If the albumen is thick, the yolk will be retained near the center of the egg and the yolk shadow outline will not be distinct.

Fig. 34-3. Gauge for measuring depth of air cell. (USDA)



The internal and external standards of individual eggs, by USDA grade, are listed below:

- *AA Quality.* The shell must be clean, unbroken, and practically normal. The air cell must not exceed 1/8 inch in depth, may show unlimited movement, and may be free or bubbly. The white must be clear and firm so that the yolk is only slightly defined when the egg is twirled before the candling light. The yolk must be practically free from apparent defects.
- *A Quality.* The shell must be clean, unbroken, and practically normal. The air cell must not exceed 3/16 inch in depth, may show unlimited movement, and may be free or bubbly. The white must be clear and at least reasonably firm so that the yolk outline is only fairly well defined when the egg is twirled before the candling light. The yolk must be practically free from apparent defects.
- *B Quality.* The shell must be unbroken and may be slightly abnormal and may show slight stains but no adhering dirt *provided* that it does not appreciably detract from the appearance of the egg. When the stain is localized, approximately $\frac{1}{32}$ of the shell surface may be slightly stained, and when the slightly stained areas are scattered, approximately $\frac{1}{16}$ of the shell surface may be slightly stained. The air cell must not exceed $\frac{3}{8}$ inch in depth, may show unlimited movement, and may be free or bubbly. The white must be clear and may be slightly weak so that the yolk outline is well defined when the egg is twirled before the candling light. The yolk may appear slightly enlarged and slightly flattened and may show other definite, but not serious, defects.

Eggs that are dirty or unsound are not eligible for grading, but are classed according to the following descriptions.

- *Dirty.* The shell must be unbroken with adhering dirt, or foreign material, prominent stains, or moderate stains covering more than one quarter of the shell surface.
- *Check.* An individual egg that has a broken shell or crack in the shell but has its shell membranes intact and its contents do not leak.
- *Leaker.* An individual egg that has a crack or break in the shell and shell membranes to the extent that the egg contents are exuding or free to exude through the shell.

When eggs are grouped, as in 1- or 3-dozen cartons, 30-dozen cases, or larger lots, reasonable tolerances for each grade are justified. Grades of individual eggs may also drop to the next grade during transport or storage, though the storage may be brief. See Table 34-3.

A grade label is shown in Fig. 34-4. Figure 34-5 shows the "broken out" appearance of eggs of the three grades.

Table 34-3

TOLERANCES FOR U.S. CONSUMER GRADES OF SHELL EGGS AT THE DESTINATION

| *U.S. consumer grade (destination)* | *Quality required*[a] | *Tolerance permitted* |
|---|---|---|
| Grade AA | 80 percent AA | Up to 20 percent A or B<br>Not over 5 percent checks |
| Grade A | 80 percent A or better | Up to 20 percent B<br>Not over 5 percent checks |
| Grade B | 80 percent B or better | Not over 10 percent checks |

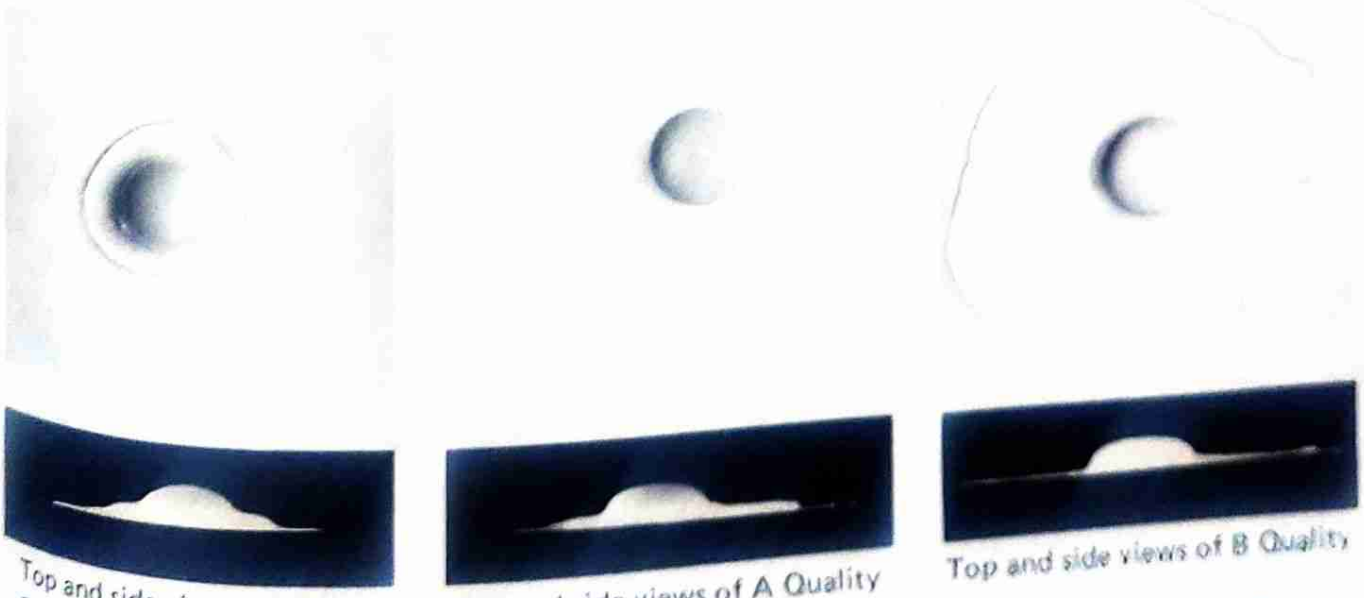
Source: FSQS, USDA, *Agricultural Handbook No. 75*; 1972 and *Federal Register*, August 4, 1981.

[a]In lots of two or more cases or cartons.




Fig. 34-4. Official grade label for one-dozen cartons. (USDA)

Fig. 34-5. The pictures on this chart show the interior quality of eggs that meet the Specification of Individual Shell Eggs with respect to albumen and yolk quality for AA Quality, A Quality, and B Quality. (USDA)



Top and side views of AA or Fresh Fancy Quality eggs

Top and side views of A Quality eggs

Top and side views of B Quality eggs

## 34.3 INHERITANCE AND EGG QUALITY

Heritabilities for major egg quality traits average about .60 for shape and weight and .35 for albumen quality and specific gravity. Presence or absence of yolk abnormalities is also inherited to a significant extent.

Because of the economic importance and relatively high heritabilities of these traits, breeders expend much effort in measuring these traits in experimental lines and varieties.

As discussed in earlier chapters, high heritability is associated with high repeatability – the bird tends to demonstrate the same trait in successive eggs, in this case. Repeatabilities of .53 for shell thickness, .66 for specific gravity, and .78 for shear press score (forced required to break the shell) support the heritability estimates.

## 34.4 EFFECT OF SEASON AND AGE OF HEN ON EGG QUALITY

Shell thickness and strength decline as laying period progresses and birds get older. This contributes to lower interior quality, especially after storage, and to increased percentages of cracked eggs. Since 25 to 40 percent of the calcium used in shell formation is drawn from skeletal stores rather than from the ration, the body may act to conserve skeletal calcium as the laying season progresses.

Cold temperatures and low relative humidity tend to cause thicker, denser shells; temperatures above 80° F and high humidity result in thinner shells, and egg size may be slightly reduced. Yolk mottling and blood spots tend to increase through the laying season and as warmer weather is reached, especially during the first year of lay.

With essentially all birds confined and receiving dry, mixed rations, there is little seasonal effect on yolk color. With birds on open range, yolks are much darker in the spring as birds eat green forages high in the xanthophylls.

## 34.5 EFFECT OF HEALTH AND NUTRITION OF THE HEN ON EGG QUALITY

All nutrients are important for egg production, but certain of them are more critical than others – that is, they are more likely to be deficient in natural rations and so are of greater concern in ration formulation. A hen uses about 2.2 grams of calcium per egg, has additional needs for other body functions, and absorption from the digestive tract is less than 100 percent. A ration level of about 3.5 percent calcium is normally recommended, though higher levels are sometimes needed in hot weather when feed consumption is lower. When more than 6 percent calcium is used, production generally declines. Relationship to and amount of vitamin $D_3$ is also important.

Certain ration adjustments, such as ratio of sodium or potassium to chloride* or additions of sodium bicarbonate, zinc propionate, calcium lactate, ascorbic acid, magnesium sulfate, calcium gluconate, and aspirin have been used in attempts to improve shell quality, but results have not been consistent.

Yolk pigmentation is subject to many nutritional influences. Xanthophylls are the main pigment, and the xanthophylls of yellow corn appear equally potent to the xanthophylls in alfalfa products in contributing to rich yolk color. Chlorophyll derivatives at relatively high concentrations have been known to cause greenish yolks.

Mottled yolks have resulted from feeding certain estrogens and from the use of cottonseed meal in layer rations. Some cottonseed meal contains gossypol, which causes an olive green or olive brown color in the yolk and in some instances, during egg storage, a mottling of the yolk color.

Albumen quality or chemical content appears to be little influenced by chemical composition of the ration. Much research has been directed to causing thicker albumen. Ammonium chloride in the ration does cause thicker albumen but results in lower production and other undesirable side effects.

High concentrations of ammonia in the atmosphere of the laying hens or egg storage area have been shown to decrease the proportion of thick albumen and to increase yolk mottling.

Some diseases affect egg quality, as well as quantity, even after the birds appear to have recovered. Birds that have had bronchitis tend to produce thin-shelled and abnormally shaped eggs. This disease and certain others may also cause thin, watery whites. Certain drugs will also affect egg quality.

Birds maintained in a controlled environment and fed a standard, well-balanced ration will generally produce eggs with uniform and consistent quality characteristics, including yolk color, albumen thickness, and flavor.

## 34.6 SHELL EGG HANDLING AND PACKAGING

Quality requirements for Grade A or AA eggs dictate that cooling facilities exist at the production unit. The quicker egg temperature is lowered to 60° F or below, the less deterioration in internal quality results. The relative humidity of storage coolers is held at 70 to 85 percent also to prevent deterioration and weight loss.

In many modern units, eggs are washed, scanned, sized, and packaged "on line" as they move by conveyor belt from the layer house and before being placed in a cooler.

Eggs may be washed in water with detergents and sanitizers. If the washing solution is warmer than the egg, the solution is not absorbed into

*R.M.G. Hamilton and B.K. Thompson, *Poultry Sci.* 59 (1980): 1294.

the egg. Such washing not only removes stains and dirt, but also reduces bacterial population.

After eggs are washed they are usually sprayed with an extremely light oil or oil emulsion, sometimes a special silicone-type product is used. This is to close pores in the shell and prevent further evaporation and deterioration; it also improves appearance.

The eggs are then sorted by size and cased or packaged in retail containers. Some suggest placing cases or cartons in the cooler for up to 24 hours prior to filling. Hot and dry cartons or cases can absorb up to a pound of water from 30 dozen eggs. Because of the weight loss, this could lower the size classification of a case or lot.

In handling eggs, the critical factors are rapid cooling, washing, oiling, sizing, packaging, and rapid shipment to the retailer, with handling restricted to as few times as possible.

Recent research in Texas has demonstrated that heavily oiling freshly laid eggs will prevent quality decline for several days in the absence of refrigeration, even with high environmental temperatures.*

## 34.7 SHELL EGG DISTRIBUTION AND MERCHANDISING

Large egg producers in densely populated areas often sell directly to retail stores, delivering sized, packaged eggs to the stores six days a week. Price, volume, and other terms are usually established by contract; the contract may provide price fluctuations according to prices that exist at an agreed-upon market.

Others sell to wholesalers who may perform some of the washing, sizing, or packaging operations and who deliver to retailers. In these cases, contracts between producer and packager-wholesaler are common but not universal. Depending on egg size, quality, volume, and consistency of production, contracts usually provide the producer a few cents premium per dozen above the market price at an agreed-upon point.

In some cases the packager-wholesaler is also a supplier of inputs – feed, day-old pullets, sanitizing materials, and so forth – and these may all be covered in the single contract. Or, the producer may provide only the building, equipment, and labor, not own the hens, and receive payment according to the number of eggs produced. Many variations in business arrangements exist. See Section 10.6, Price Protection.

Those who produce such a small number of eggs that daily pickup is not justified usually sell to a local assembler on the basis of "current receipts." This term refers to the local price being paid currently for eggs, based on competition among such assemblers, and the price they can obtain from retailers, wholesalers, or processors for eggs of the various sizes and grades their operations yield.

*F.A. Gardner et al., *Poultry Sci.* 59 (1980): 1612.

In recent years there has generally been a 10- to 15-cent-per-dozen differential between the "current receipts" price and average price received by consistent producers of high quality and large volumes of eggs sold under marketing agreement. As with other commodities, efficiencies in handling larger and uniform quantities contribute to the price differential.

## 34.8 EGG PROCESSING AND PRODUCTS

The 10 to 12 percent of the eggs not used each year as "shell eggs" are processed for use in prepared food items, such as cake mixes, pastries, eggnog flavoring beads, chiffon pies, egg rolls, noodles, mayonnaise, and other food products. Liquid eggs can be frozen or dried and later used for scrambled eggs and other purposes. Whites and yolks may be used separately.

Inspection requirements provided by the Egg Products Inspection Act of 1970 specify that the quality of eggs used must be designated, that rigid sanitation practices must be followed, that all products except dried egg whites must be pasteurized, and that all products in all stages of processing be subject to inspection. The standard handling of liquid egg products at 140°F for 3.5 minutes also reduces foaming properties of the whites. Dried egg whites not pasteurized must be held, after being dried for at least seven days at a minimum temperature of 130°F.

## 34.9 CONSUMPTION TRENDS

Per capita egg consumption declined steadily from 1960 to 1977, and since then has increased slightly. See also Fig. 34-6. Possible contributors to the decline prior to 1977 were an increase in the number who eat little or no breakfast, and studies relating cholesterol to heart disease.

Future trends in egg consumption will likely depend on development of new uses, new packaging materials, and advances in preservation techniques. The egg roll, four large Grade A eggs cooked in a polypropylene tube, provides uniform egg slices for salads, hors d'oeuvres, or to dress up other foods on the serving table. It has long shelf life, enduring flavor, and has gained quick popularity in restaurants and institutions. The breakfast trade in fast-food outlets may increase egg consumption (Fig. 34-7).

An eggnog mix that dissolves instantly in milk, frozen french toast, and a frozen, ready-to-cook egg omelet for a main course or sandwich filling are other illustrations.

Increasing shelf life of eggs for home use, perhaps by freezing or by replacing the shell with a sealed plastic cup, are possibilities.

The American Egg Board and other industry organizations provide consumer education and research efforts aimed at increasing egg consumption and utilization.